

Who spoke in the group videos?

Final comparison on 11 recordings

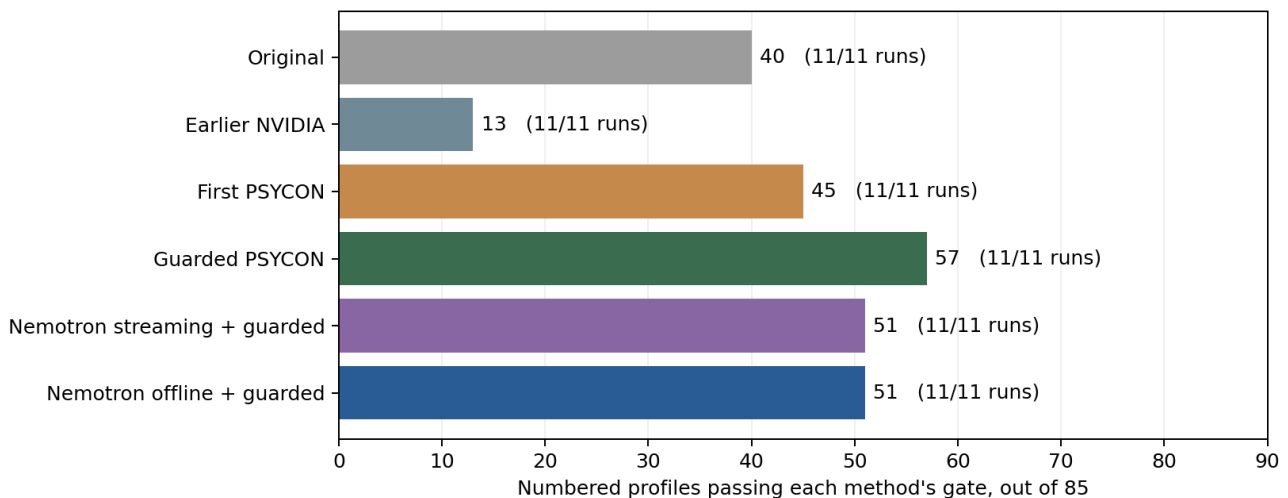
1 October 2026

We tested all 11 recordings with the same source audio and numbered faces. They contain 85 marked places across the videos. These are person-recording slots, since someone may appear in more than one recording. Nemotron offline produced 51 ready profiles, while guarded PSYCON produced 57. PSYCON therefore remains our main source. Nemotron stays in the benchmarks.

Method	Ready out of 85	Usable seconds
Original	40	395.14
Earlier NVIDIA	13	524.92
First PSYCON	45	4,364.61
Guarded PSYCON	57	4,640.04
Nemotron streaming with current checks	51	4,849.58
Nemotron offline with current checks	51	5,179.03

How we did it

1. We extracted shared 16 kHz audio and numbered the visible faces.
2. Community-1 or Nemotron found anonymous speaker turns and detected overlap.
3. TalkNet checked the visible speaker. SpeechBrain compared the voice across separate turns. Repeated, consistent evidence linked a voice to a face.
4. A profile needed three usable seconds and a valid voice embedding. We removed detected overlap, uncertain boundaries, and nonusable audio windows from acoustic measurements. Playback files came from the original audio.



All methods stay in the benchmark, even when they produce fewer profiles.

What changed for Nemotron

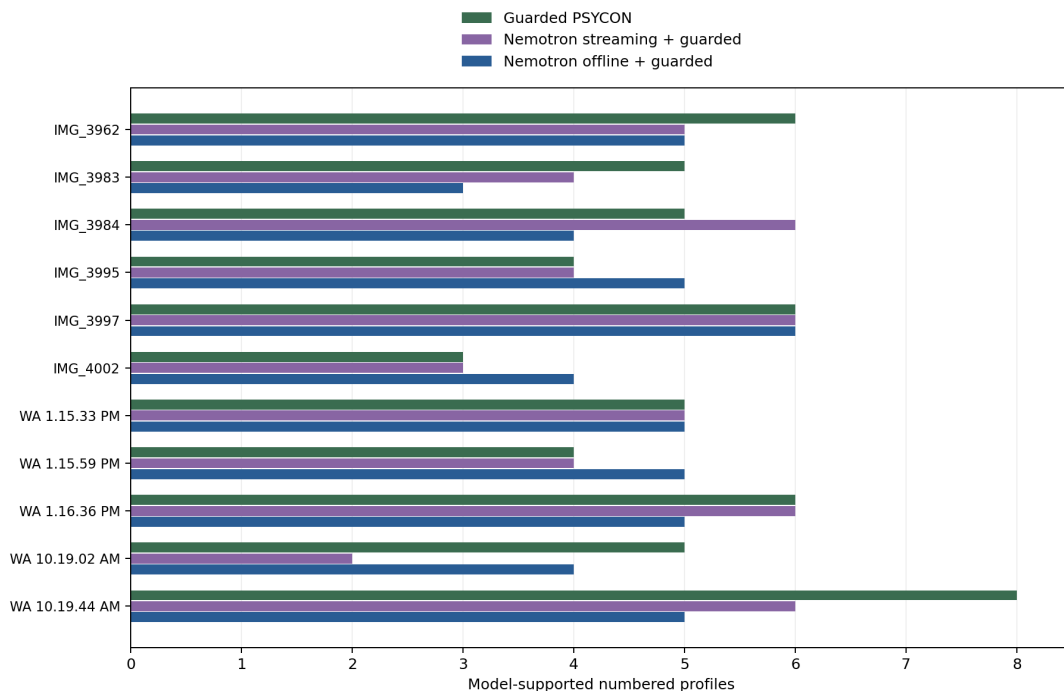
The teammate uses the offline preset. It processes 27.2 seconds at a time and looks ahead by 3.2 seconds. Our earlier NVIDIA mode used 0.72-second chunks and 0.32 seconds of lookahead. Both keep speaker history, but offline gets more context. We added that preset with pinned weights and the current guarded matcher.

The streaming control produced 51 profiles. It reused the earlier NVIDIA turns with the same face and voice checks as PSYCON. The offline run used those checks too. The offline setting matched that control's coverage. Offline retained 5,179.03 usable seconds, compared with 4,640.04 from PSYCON. It gave us more audio across fewer participants, but this does not prove better identity accuracy.

Earlier NVIDIA used 80 anonymous channels across the videos. Its 13 ready profiles were face-linked outputs. The teammate's profile counter uses total channel speech, including overlap, and its face mapping is manual. Its repository has no saved results for these recordings.

Recording	Faces	Original	Earlier NV	First PS	Guarded PS	Stream+	Offline+
IMG_3962	8	5	1	5	6	5	5
IMG_3983	8	1	0	4	5	4	3
IMG_3984	7	4	1	4	5	6	4
IMG_3995	8	4	0	2	4	4	5
IMG_3997	7	3	3	5	6	6	6
IMG_4002	8	5	1	2	3	3	4
WA 1.15.33 PM	8	5	1	4	5	5	5
WA 1.15.59 PM	8	3	2	3	4	4	5
WA 1.16.36 PM	7	3	0	3	6	6	5
WA 10.19.02 AM	8	2	1	5	5	2	4
WA 10.19.44 AM	8	5	3	8	8	6	5
Total	85	40	13	45	57	51	51

WA means a WhatsApp filename. Stream+ and Offline+ use the same guarded checks as PSYCON. The earlier methods used their own gates. Counts measure coverage, not identity accuracy.



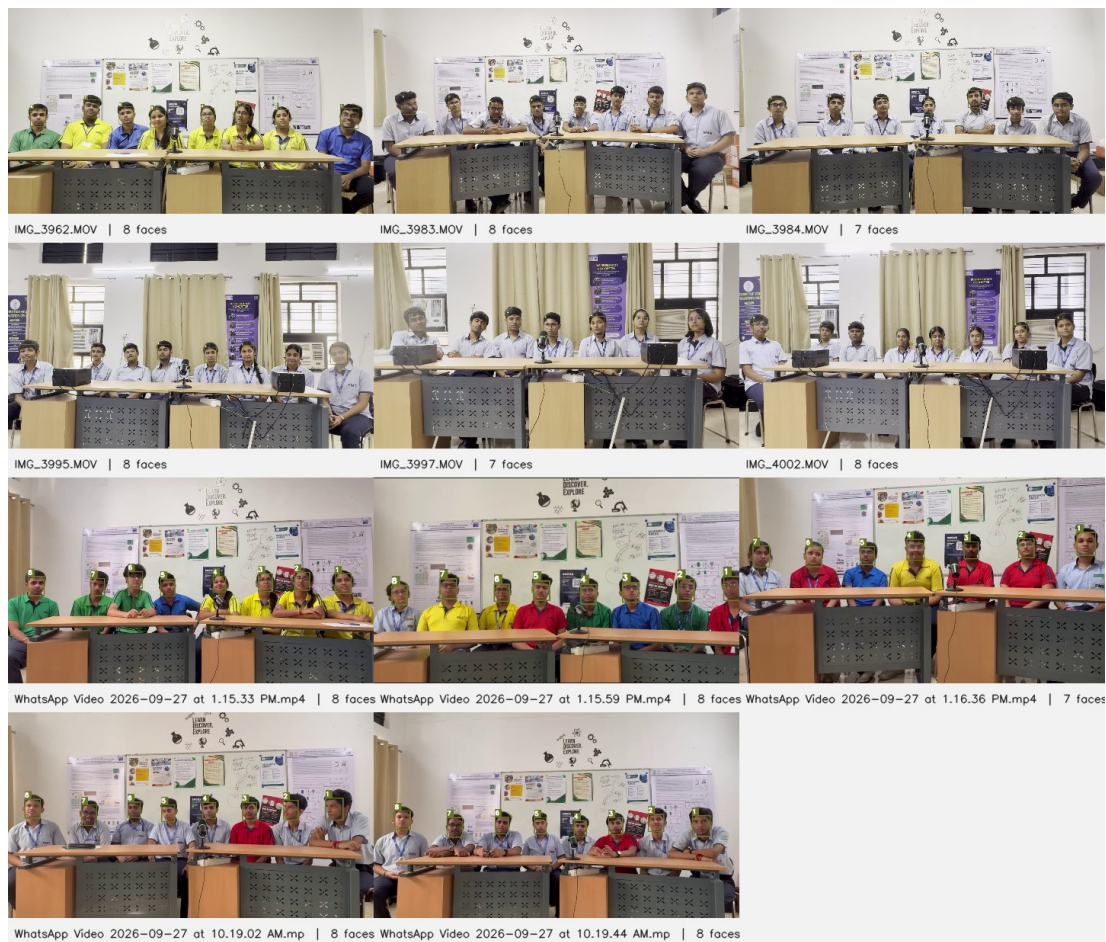
What remains uncertain

Guarded PSYCON still leaves 28 slots incomplete. All have zero assigned clean speech. Seventeen have tentative visual candidates for review, and eleven have none. This does not prove those people were silent. 1420.81 seconds of clean speech remain unlinked to a face, and the model detected 472.54 seconds of overlap.

Mixed original audio can appear in playback, but overlap stays out of a single person's acoustic profile. Nine PSYCON slots have no person-specific playback clip. Three videos have only seven marked faces. An unmarked eighth person cannot receive a numbered face profile from those markings. A sustained vowel measurement is unavailable because the videos lack that task.

These are model-supported assignments. Most have no independent human labels. More profiles do not establish better accuracy. Psychological predictions have not been validated. Training continues to read ready PSYCON profiles; the Nemotron benchmark has a separate schema.

Offline Nemotron found 4 slots PSYCON left incomplete, but lost 10 of PSYCON's ready slots. 5 assigned intervals disagreed with saved PSYCON face observations. These clips need inspection; the check does not tell us which model is correct. The results do not prove that PSYCON has reached the maximum recoverable coverage.



The marked faces from all 11 recordings. Numbers belong to each recording.

All 66 recording-method runs finished. All 243 automated tests passed across the host and Docker runtimes, including real models, ffmpeg, raw WESAD S2, and live processing/export. Artifact validation passed with zero errors across 377 playback and review clips. It checked source intervals, profile gates, and audio timing. The PDF pages were checked after compilation.

Sources: teammate benchmark code, official Nemotron model card. The project includes the full benchmark summary and rejection audit.